

SUPPLEMENTAL MATERIAL

Comparison of Methods to Demonstrate Bioequivalence in Case of High Variability and Group Heterogeneity

Annabelle Walz, Daniel Kaschek, IntiQuan AG, Basel, Switzerland

Brandon Greene, Sabine Pestel, CSL Behring Innovation GmbH, Marburg, Germany

IntiQuan⁺
A ProductLifeGroup Company

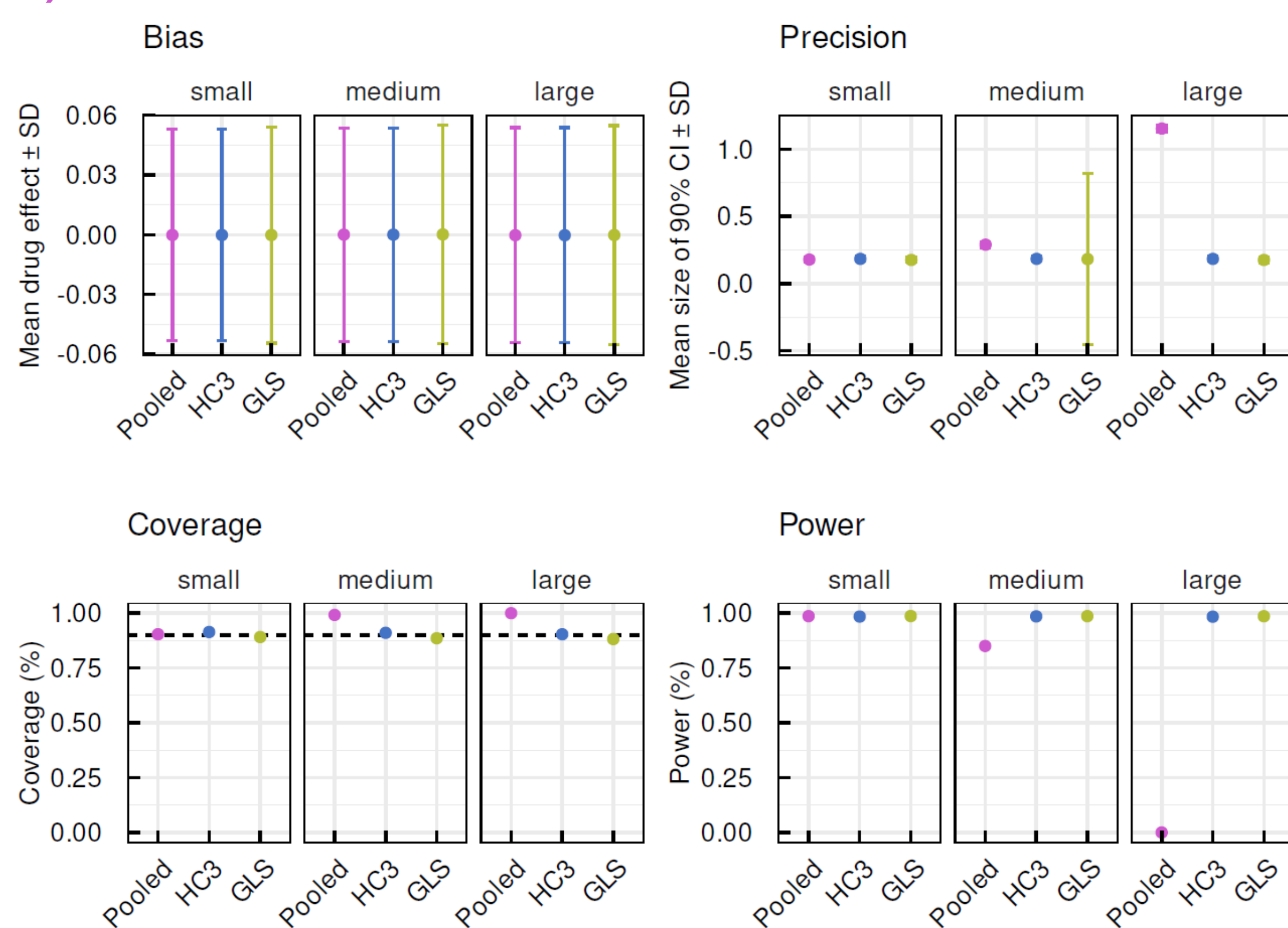
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Fig. 1 | Equal variances between groups

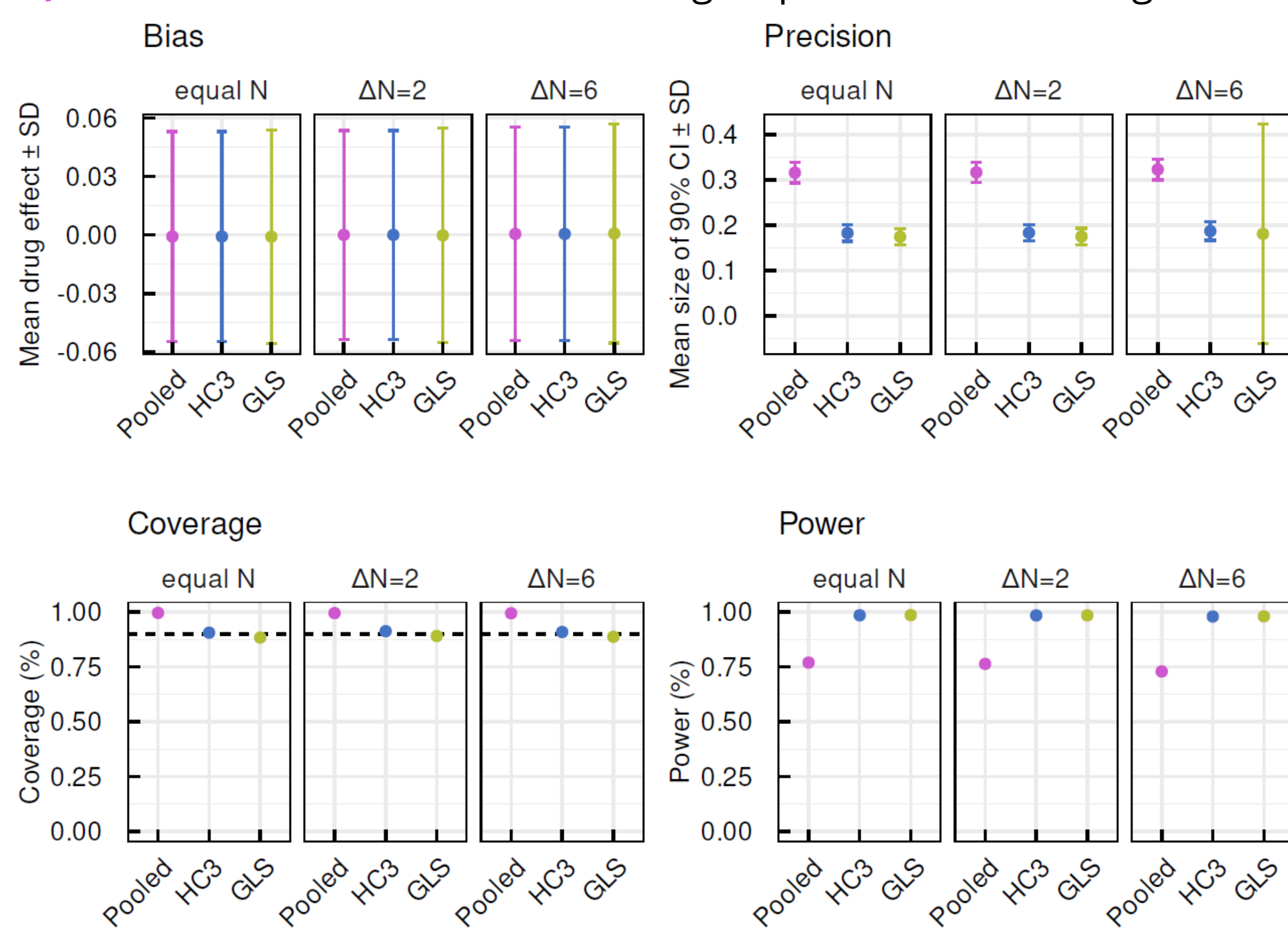
In these simulations, a sex effect on AUC and equal variances among groups (across drugs and sex) were assumed.

- Overall, HC3 and GLS performed equally well. Both outperformed the standard BE method (Pooled).
- Their benefit over the standard BE method (Pooled) increased with increasing sex effect size **(A)**.
- Their benefit over the standard BE method (Pooled) remained stable with increasing imbalance between group size (drug or sex) **(B, C)**.

A) SEX EFFECT SIZE Different size of sex effect



B) SAMPLE SIZE BY DRUG Different group size between drugs



C) SAMPLE SIZE BY SEX Equal or higher group size in ♀ compared to ♂

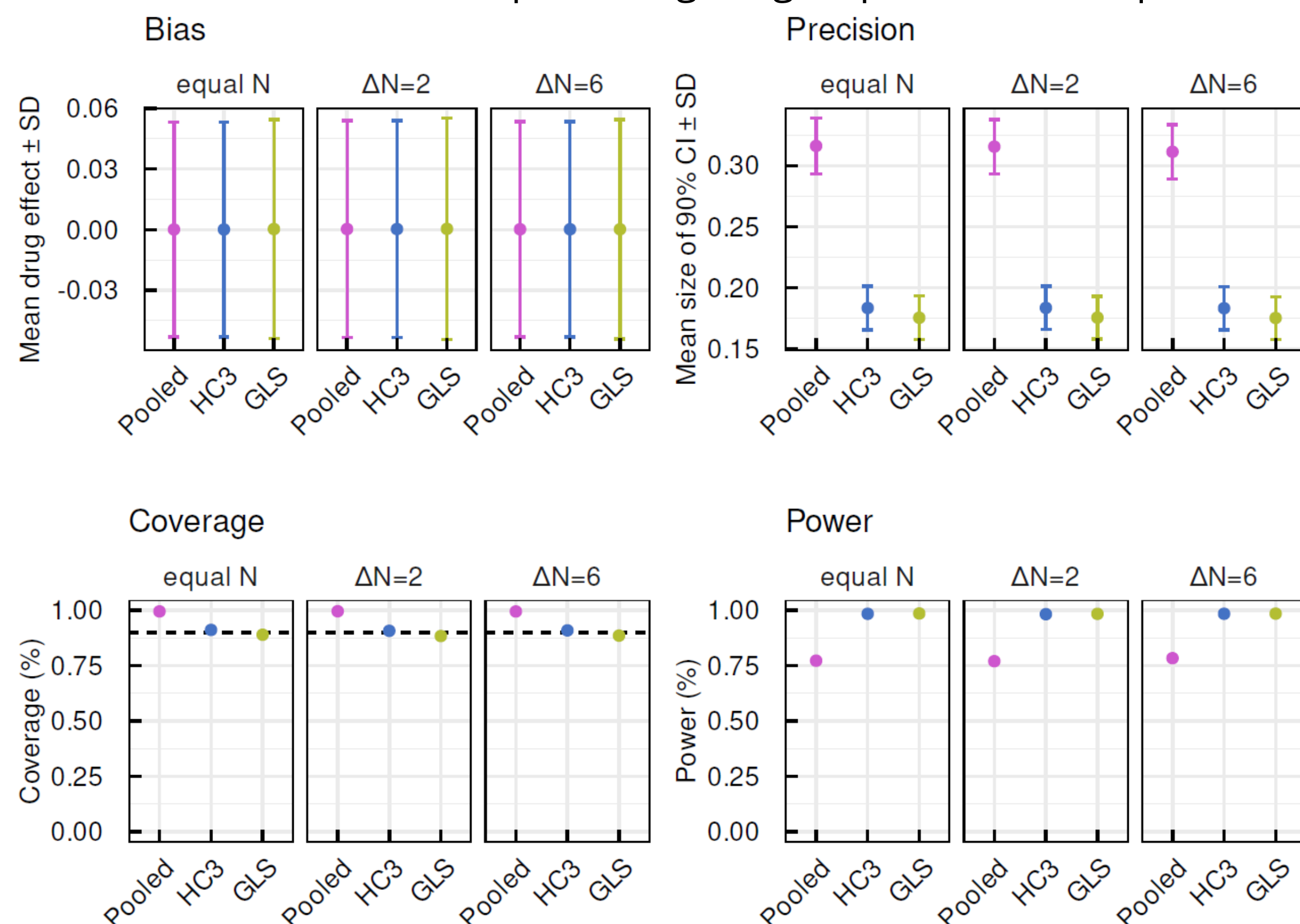
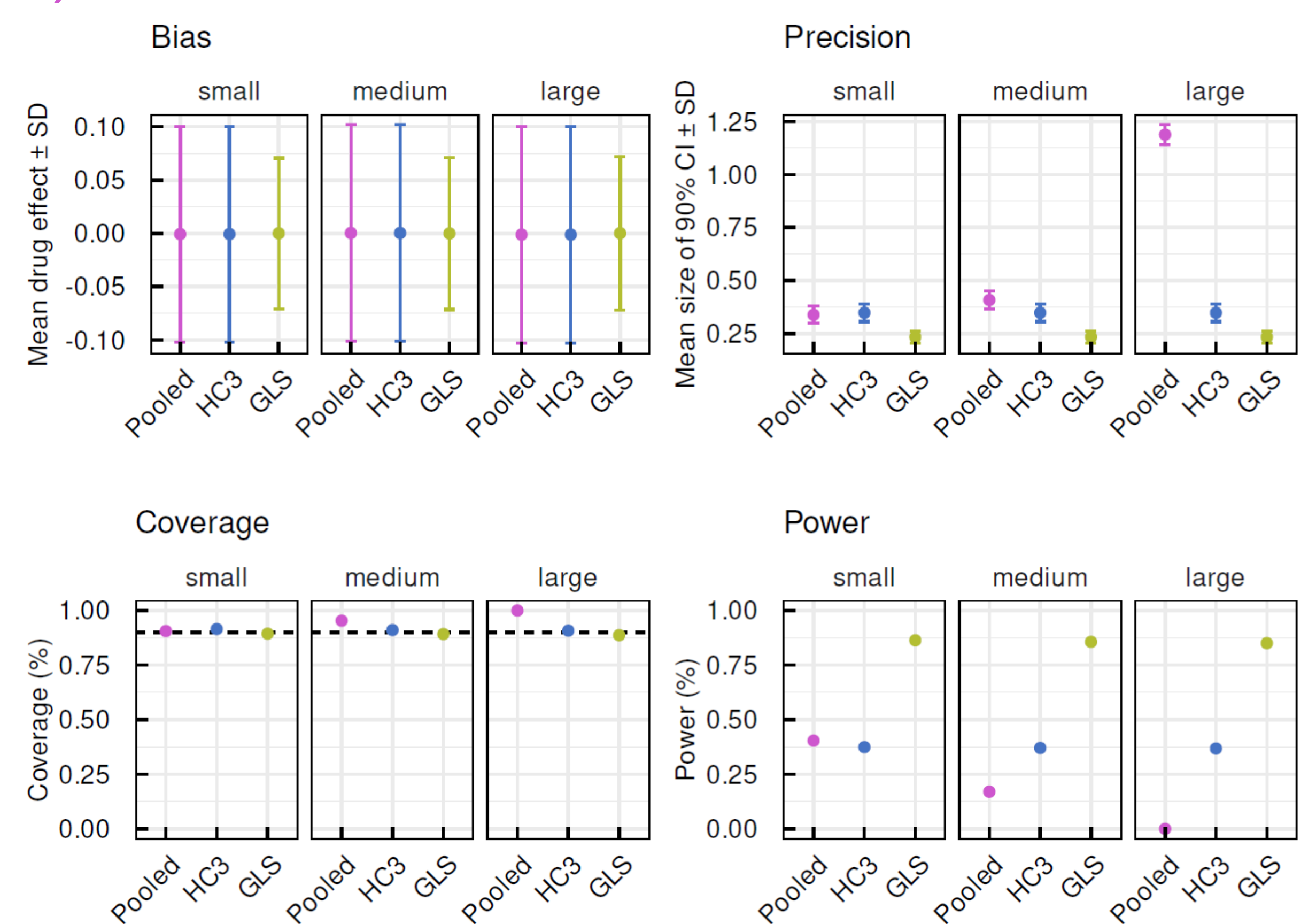


Fig. 2 | Unequal variances between groups

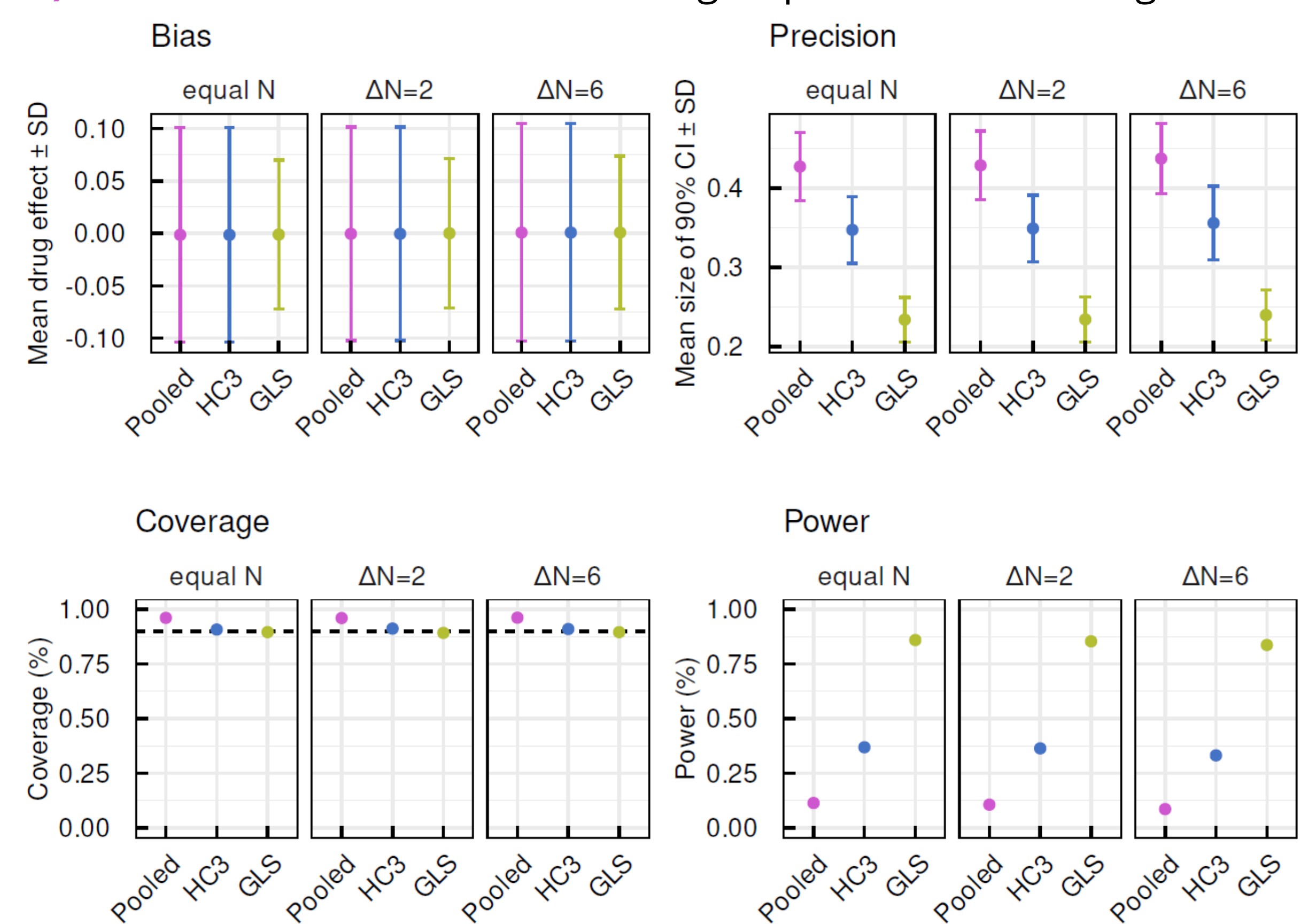
In these simulations, a sex effect on AUC and unequal variances among groups (males have 2.5-fold higher SD than females) were assumed.

- Overall, GLS outperformed HC3 and the standard BE method (Pooled).
 - The benefit of GLS (and HC3) increased with increasing sex effect size **(A)**.
- The highest power was achieved with
 - small effect size **(A)**, and
 - equal group size across drugs **(B)**, and
 - when the sex ratio shifted towards the sex with lower variability (i.e., more females than males) **(C)**.

A) SEX EFFECT SIZE Different size of sex effect



B) SAMPLE SIZE BY DRUG Different group size between drugs



C) SAMPLE SIZE BY SEX Equal or higher group size in ♀ compared to ♂

